

CN HW 6

-group 31

- 1) The time taken to send the file with the sack enabled was **2:37.35**

Attaching screenshot below

```
net.ipv4.tcp_sack = 1
dhruv@dhruv-Surface-Pro:~/Documents/CN_HW6$ /usr/bin/time scp /home/dhruv/Documents/CN_HW6/big.zip Bhuvan@192.168.43.219:/home/Bhuvan/
Bhuvan@192.168.43.219's password:
big.zip                                100% 1176MB  7.8MB/s   02:30
4.15user 2.16system 2:37.35elapsed 4%CPU (0avgtext+0avgdata 12644maxresident)k
0inputs+0outputs (0major+17229minor)pagefaults 0swaps
```

- 2) The time taken to send the file with the sack enabled was **3:21.05**

With Sack enabled the sender only has to send the missing files as identified therefore without sack it takes longer

Attaching screenshot below

```
dhruv@dhruv-Surface-Pro:~/Documents/CN_HW6$ /usr/bin/time scp /home/dhruv/Documents/CN_HW6/big.zip Bhuvan@192.168.43.219:/home/Bhuvan/
Bhuvan@192.168.43.219's password:
big.zip                                100% 1176MB  6.6MB/s   02:58
4.32user 2.29system 3:21.05elapsed 3%CPU (0avgtext+0avgdata 13420maxresident)k
0inputs+0outputs (0major+24240minor)pagefaults 0swaps
dhruv@dhruv-Surface-Pro:~/Documents/CN_HW6$ /usr/bin/time scp /home/dhruv/Documents/CN_HW6/big.zip Bhuvan@192.168.43.219:/home/Bhuvan/
```

- 3) The total number of SACKS received 94088 with tcp_sack enabled

- 4) The number of ranges were: 1: 90852 2: 3151 3:86 >4:0

```
Option kind=8^C
dhruv@dhruv-Surface-Pro:~/Documents/CN_HW6$ grep '^SACK:' sack_enabled.log | wc
-l
94088
dhruv@dhruv-Surface-Pro:~/Documents/CN_HW6$ grep '^SACK:' sack_enabled.log |
awk '{n=gsub(/\[/, "[" ); if(n==1)c1++; else if(n==2)c2++; else if(n==3)c3++; else
c4++;}
END{printf("ranges=1:%d 2:%d 3:%d ≥4:%d\n",c1,c2,c3,c4)}'
ranges=1:90852 2:3151 3:86 ≥4:0
dhruv@dhruv-Surface-Pro:~/Documents/CN_HW6$
```

5)The other kinds were: 2,3,4,8

2: This gives us the MSS which is the maximum segment size essentially the largest TCP payload the sender is willing to receive .

3:It is used to scale the Window after tcp handshake. Contains scale value and the window size is scaled by $2^{\text{(the value sent)}}$. Especially useful for large transfers

4: This implies that that SACKS are permitted. It does not carry SACK blocks it just lets us know they are allowed for future use.

8:This carries the timestamp for each segment allows us to calculate RTT without any errors.

Attaching a Log file in the zip with a small chunk of the log file created during send of 1.2 GB file with tcp_sack enabled.